

Markscheme

Mathematics: analysis and approaches
Practice question bank

78. (a)

The revenue of a company was \$50 000 in year 1. Revenue increased by \$4000 each year for the first 5 years. From year 6 onward, revenue instead grows by 8% each year (compared to the previous year).

Find the revenue of the company in year 8, correct to the nearest dollar.

find the year-5 revenue using the arithmetic phase: $50\,000 + 4(4000) = 66\,000$ (M1) A1

from year 6, model with a geometric phase based on this value: $66\,000(1.08)^n$ where n counts years after year 5 (M1)

year 8 corresponds to $n = 3$: $66\,000(1.08)^3$ (M1)

$\approx \$83\,141$ A1

[6 marks]